

# A PRIME-REPRESENTING FUNCTION

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A function  $f(x)$  is said to be a prime-representing function if  $f(x)$  is a prime number for all positive integral values of  $x$ . It will be shown that there exists a real number  $A$  such that  $[A^{3^x}]$  is a prime-representing function, where  $[R]$  denotes the greatest integer less than or equal to  $R$ .

Let  $p_n$  denote the  $n$ th prime number. A. E. Ingham<sup>1</sup> has shown that

$$(1) \quad p_{n+1} - p_n < K p_n^{5/8}$$

where  $K$  is a fixed positive integer.

LEMMA. *If  $N$  is an integer greater than  $K^3$  there exists a prime  $p$  such that  $N^3 < p < (N+1)^3 - 1$ .*

PROOF. Let  $p_n$  be the greatest prime less than  $N^3$ . Then

$$(2) \quad N^3 < p_{n+1} < p_n + K p_n^{5/8} < N^3 + K N^{15/8} < N^3 + N^2 < (N+1)^3 - 1.$$

Let  $P_0$  be a prime greater than  $K^3$ . Then by the lemma we can construct an infinite sequence of primes,  $P_0, P_1, P_2, \dots$ , such that  $P_n^3 < P_{n+1} < (P_n + 1)^3 - 1$ . Let

$$(3) \quad u_n = P_n^{3^{-n}}, \quad v_n = (P_n + 1)^{3^{-n}}.$$

Then

$$(4) \quad v_n > u_n, \quad u_{n+1} = P_{n+1}^{3^{-n-1}} > P_n^{3^{-n}} = u_n,$$

$$(5) \quad v_{n+1} = (P_{n+1} + 1)^{3^{-n-1}} < (P_n + 1)^{3^{-n}} = v_n.$$

It follows at once that the  $u_n$  form a bounded monotone increasing sequence. Let  $A = \lim_{n \rightarrow \infty} u_n$ .

THEOREM.  $[A^{3^x}]$  is a prime-representing function.

PROOF. From (4) and (5) it follows that  $u_n < A < v_n$ , or  $P_n < A^{3^n} < P_n + 1$ . Therefore  $[A^{3^n}] = P_n$  and  $[A^{3^x}]$  is a prime-representing function.

<sup>1</sup> A. E. Ingham, *On the difference between consecutive primes*, Quart. J. Math. Oxford Ser. vol. 8 (1937) pp. 255-266.